

Assignment - II Solutions

Astronomy & Astrophysics II

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Problem 1

The emission coefficient for thermal Bremsstrahlung is given by:

$$j_\nu(r) = \frac{32Z^2e^6}{3mc^3} \left(\frac{2\pi}{3mkT(r)} \right)^{1/2} n_e n_i \bar{g}_{ff} e^{-h\nu/kT(r)} \quad (1)$$

where m is the mass of the particle, Z is the charge of the ion, n_e and n_i are the number densities of the particles and the ions, and $\bar{g}_{ff}$ is the velocity-averaged Gaunt factor.

Using Kirchhoff's law ($j_\nu = \alpha_\nu B_\nu(T)$), the absorption coefficient is given by:

$$\alpha_\nu(r) = \frac{4Z^2e^6}{3hmc} \left(\frac{2\pi}{3mkT(r)} \right)^{1/2} n_e n_i \bar{g}_{ff} \nu^{-3} \left[1 - e^{-h\nu/kT(r)} \right] \quad (2)$$

The radiation transfer equation (RTE) along the radial direction is given by:

$$\frac{dI_\nu(r)}{dr} = j_\nu(r) - \alpha_\nu(r)I_\nu(r) \quad (3)$$

Assumptions:

- Electrons are the particles and protons are the ions, i.e., $m = 9.11 \times 10^{-28} \text{ g}$, and $Z = 1$.
- Equal number densities, i.e., $n_e = n_i \equiv 10^8 \text{ cm}^{-3}$.

With the above assumptions, the RTE is solved numerically to obtain the radial intensity profile.

(i) $T = r/r_0 \times 10^4 \text{ K}$

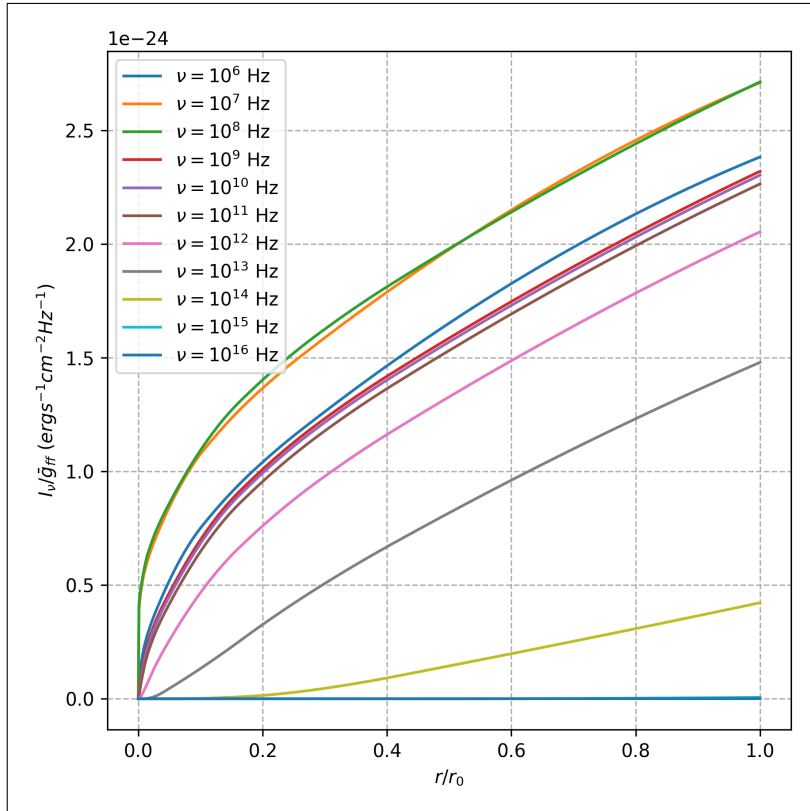


Figure 1: Radial intensity profile in frequency band $10^6 - 10^{16} \text{ Hz}$

(ii) $T = e^{-r/r_0} \times 10^6 \text{ K}$

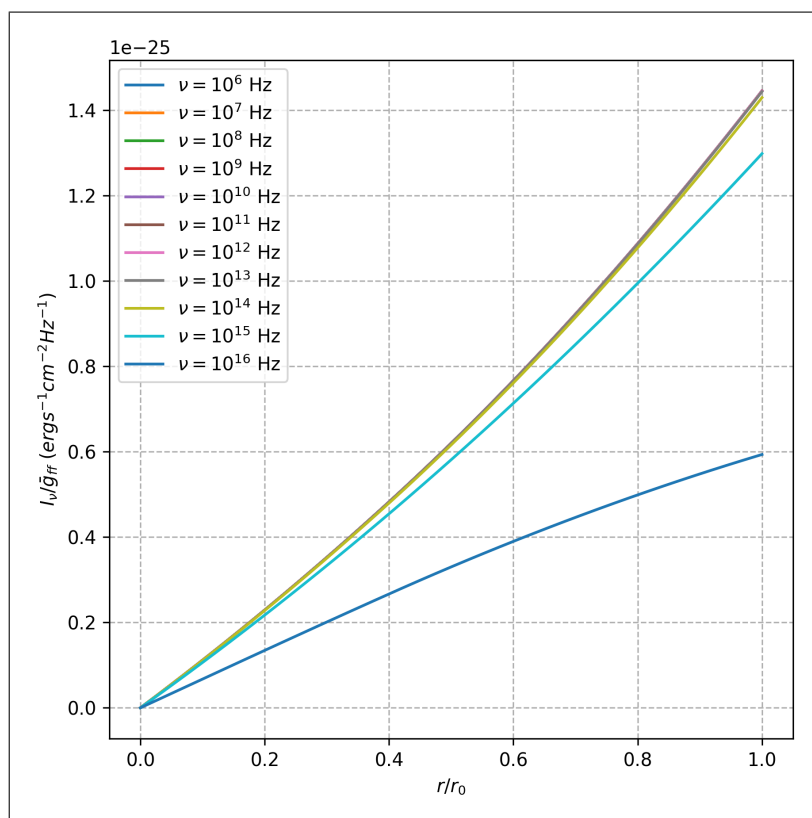


Figure 2: Radial intensity profile in frequency band $10^6 - 10^{16} \text{ Hz}$

Links to code: [Question 1\(i\)](#), [Question 1\(ii\)](#)

Problem 2

Electric & Magnetic Fields

Consider the electron to be moving with a constant velocity v along the x -axis and the proton to be at rest at a distance b along the $-y$ axis. From the Liénard-Wiechert potentials, the electric and magnetic fields at $(0, -b, 0)$ are given by:

$$E_x(t) = -\frac{q\gamma vt}{(\gamma^2 v^2 t^2 + b^2)^{3/2}} \quad (4)$$

$$E_y(t) = \frac{q\gamma b}{(\gamma^2 v^2 t^2 + b^2)^{3/2}} \quad (5)$$

$$B_z(t) \equiv \beta E_y(t) = \frac{q\gamma \beta b}{(\gamma^2 v^2 t^2 + b^2)^{3/2}} \quad (6)$$

(i) $\mathbf{v} = 0.1\mathbf{c}$

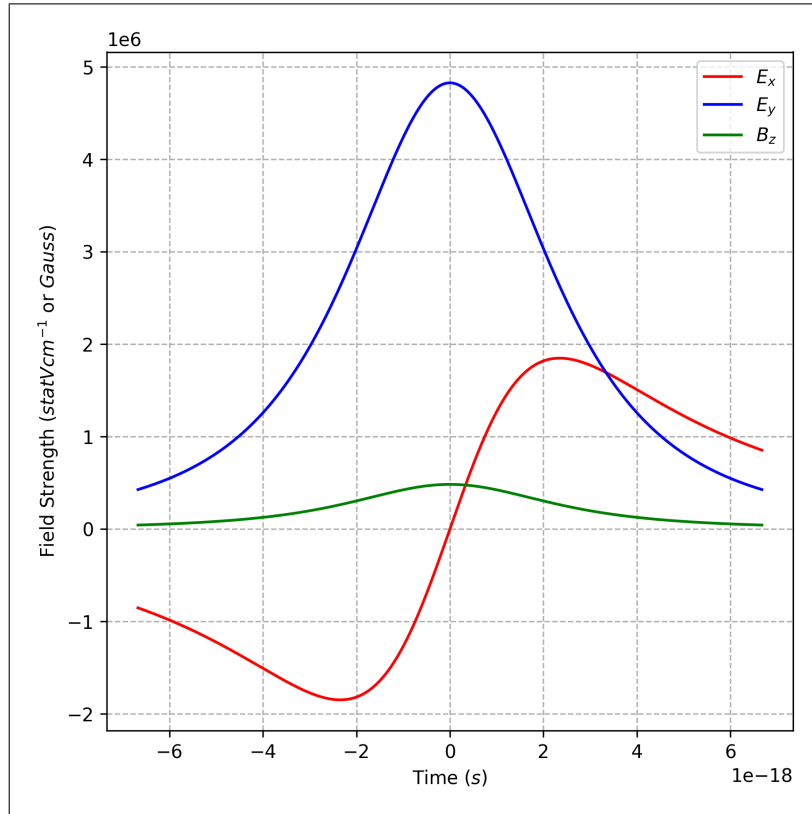


Figure 3: Electric & magnetic fields' evolution at $(0, -b, 0)$

(ii) $\mathbf{v} = 0.999\mathbf{c}$

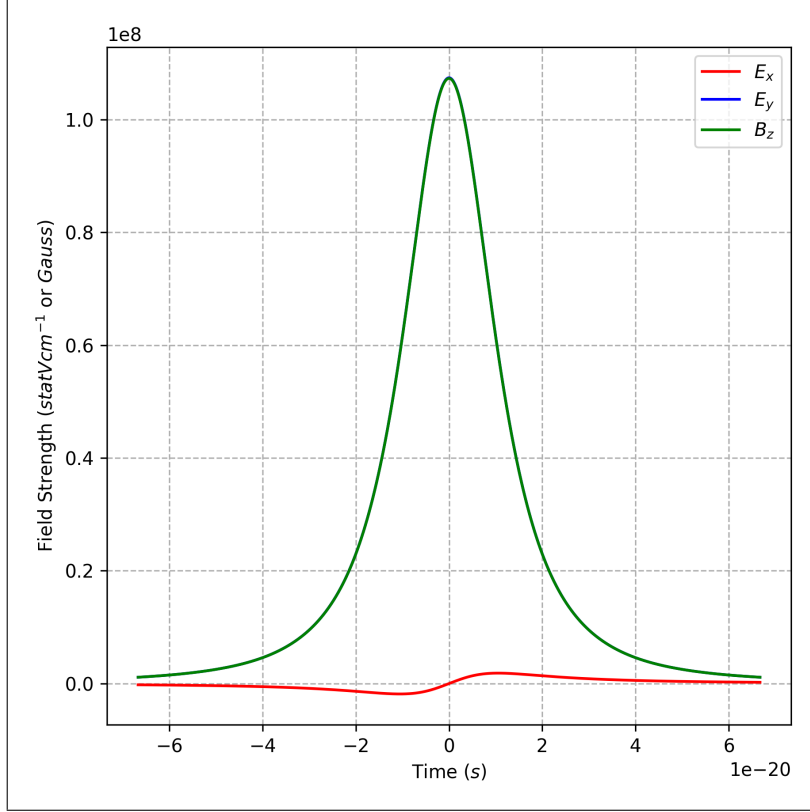


Figure 4: Electric & magnetic fields' evolution at $(0, -b, 0)$

Bremsstrahlung Emission Spectra

The emission coefficient for a **classical** free-free emission is given by:

$$j_{\nu}^{ff} = \frac{16\pi Z^2 e^6}{3\sqrt{3}m^2 c^3 v} n_e n_i g_{ff}(v, \omega) \quad (7)$$

where m and v are the mass and the velocity of the electron, Z is the charge of the proton, n_e and n_i are the number densities of the proton and the electron, and g_{ff} is the velocity & frequency dependent Gaunt factor.

Assuming $n_e = n_i \equiv 10^8 \text{ cm}^{-3}$, the Bremsstrahlung emission coefficients are given by:

- (i) $\mathbf{v} = 0.1\mathbf{c}$: $j_{\nu}^{ff} = 1.772 \times 10^{-26} g_{ff} \text{ ergs}^{-1} \text{ cm}^{-3} \text{ Hz}^{-1}$.
- (ii) $\mathbf{v} = 0.999\mathbf{c}$: $j_{\nu}^{ff} = 1.774 \times 10^{-27} g_{ff} \text{ ergs}^{-1} \text{ cm}^{-3} \text{ Hz}^{-1}$.

Since j_{ν}^{ff} scales as v^{-1} , the non-relativistic particle emits ~ 10 times more Bremsstrahlung than the relativistic particle.

Links to code: [Question 2\(i\)](#), [Question 2\(ii\)](#)

Problem 3

For a system at constant temperature T and uniform density n_e , n_p , the thermal intensity profile, assuming no incident radiation ($I_\nu(0) = 0$), is given by:

$$I_\nu(\tau_\nu) = B_\nu(T) (1 - e^{-\tau_\nu}) \quad (8)$$

where $\tau_\nu = \alpha_\nu L$ is the optical depth of the system of length L .

The free-free absorption coefficient is given by:

$$\alpha_\nu = \frac{4Z^2 e^6}{3hmc} \left(\frac{2\pi}{3mkT} \right)^{1/2} n_e n_p \nu^{-3} \bar{g}_{ff} (1 - e^{-h\nu/kT}) \quad (9)$$

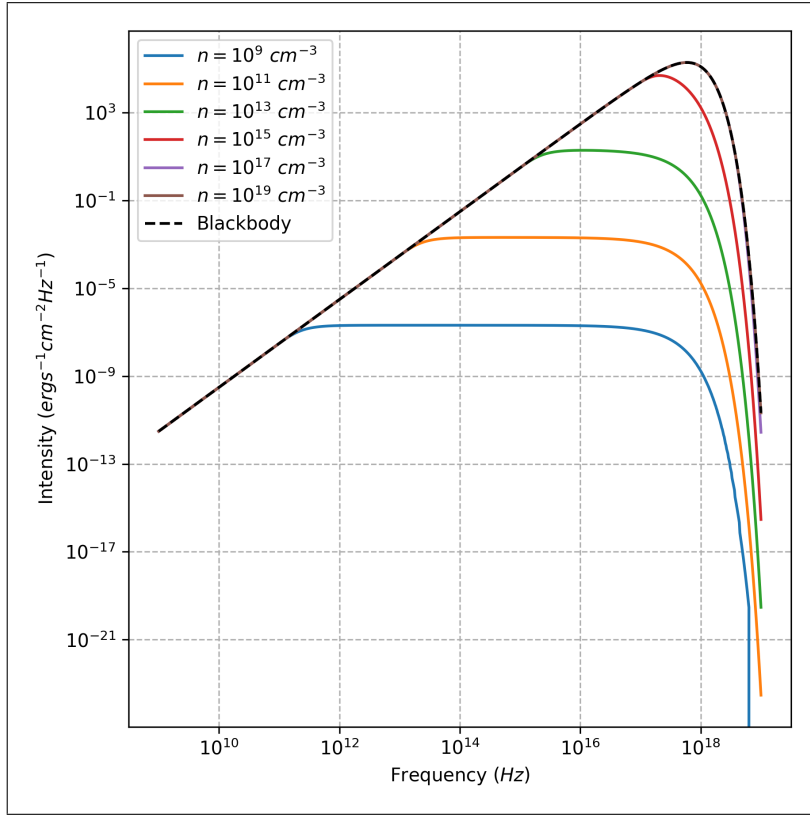


Figure 5: Thermal Bremsstrahlung spectrum at $T = 10^7$ K

The Bremsstrahlung spectrum approaches the blackbody spectra in the optically thick limit of the system. Assuming $n_e = n_p \equiv n$, in the low frequency (Rayleigh-Jeans) regime, $\alpha_\nu \propto n^2 \nu^{-2}$, and in the high frequency (Wien) regime, $\alpha_\nu \propto n^2 \nu^{-3}$. Hence, the system universally is optically thick at low frequencies, and at higher densities, it remains optically thick even at higher frequencies. Therefore, the Bremsstrahlung spectrum saturates to the blackbody limit at high densities and low frequencies.

The relative difference between Bremsstrahlung and the blackbody spectra at the same temperature T is given by:

$$\left| \frac{\Delta B_\nu(T)}{B_\nu(T)} \right| = \frac{B_\nu(T) - I_\nu}{B_\nu(T)} = e^{-\tau_\nu} \quad (10)$$

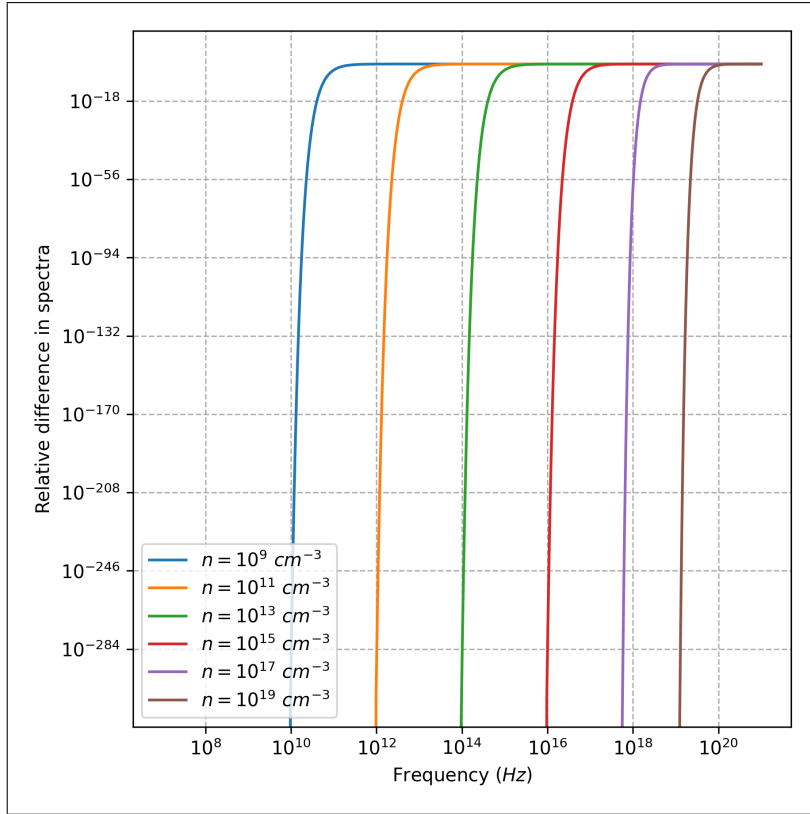


Figure 6: Relative difference between Bremsstrahlung and blackbody spectra at $T = 10^7$ K

Links to code: [Question 3\(a\)](#), [Question 3\(b\)](#)

Problem 4

(A) Emitted Power

For an electron moving with velocity $v = \beta c$ in a helical path in magnetic field B , the total power emitted via synchrotron (or cyclotron) radiation, averaged over an isotropic distribution of the velocity, is given by:

$$P_{sync} = \frac{4e^4}{9m^2c^3} \beta^2 \gamma^2 B^2 \quad (11)$$

- (i) $\beta = 0.99$, $B = 0.1$ Gauss: $\gamma = 7.089 \implies P_{sync} = 5.211 \times 10^{-16} \text{ ergs}^{-1}$
- (ii) $\beta = 0.999$, $B = 10^8$ Gauss: $\gamma = 22.366 \implies P_{sync} = 5.282 \times 10^3 \text{ ergs}^{-1}$
- (iii) $\beta = 0.1$, $B = 10^8$ Gauss: $\gamma = 1.005 \implies P_{sync} = 1.069 \times 10^{-1} \text{ ergs}^{-1}$

(B) Cooling Timescale

The cooling timescale, defined as the characteristic time required for the electron to radiate away its total energy by synchrotron (or cyclotron) emission, is given by:

$$t_{cool} \equiv \frac{E}{P_{sync}} = \frac{\gamma mc^2}{\frac{4e^4}{9m^2c^3} \beta^2 \gamma^2 B^2} = \frac{9m^3c^5}{4e^4 \beta^2 \gamma B^2} \quad (12)$$

- (i) $\beta = 0.99$, $B = 0.1$ Gauss: $\gamma = 7.089 \implies t_{cool} = 1.114 \times 10^{10} \text{ s}$
- (ii) $\beta = 0.999$, $B = 10^8$ Gauss: $\gamma = 22.366 \implies t_{cool} = 3.467 \times 10^{-9} \text{ s}$
- (iii) $\beta = 0.1$, $B = 10^8$ Gauss: $\gamma = 1.005 \implies t_{cool} = 7.699 \times 10^{-6} \text{ s}$

(C) Magnetic field strength constraint for minimal energy loss

For the energy change in one orbital period to be insignificant, the energy radiated away by the electron in one orbital period must be far less than its total energy, i.e., $\Delta E \ll E$.

The gyration frequency of the electron in the magnetic field is given by:

$$\omega_B = \frac{eB}{\gamma mc}$$

So, the orbital period is given by:

$$T_{orb} \equiv \frac{2\pi}{\omega_B} = \frac{2\pi\gamma mc}{eB}$$

The energy lost in one orbital period is given by:

$$\Delta E = P_{sync} T_{orb} = \left(\frac{4e^4}{9m^2c^3} \beta^2 \gamma^2 B^2 \right) \left(\frac{2\pi\gamma mc}{eB} \right) = \frac{8\pi e^3}{9mc^2} \beta^2 \gamma^3 B \quad (13)$$

Therefore, the constraint on the magnetic field strength is given by:

$$\begin{aligned} \frac{8\pi e^3}{9mc^2} \beta^2 \gamma^3 B &\ll \gamma mc^2 \\ \implies B &\ll \frac{9m^2c^4}{8\pi e^3 \beta^2 \gamma^2} \end{aligned} \quad (14)$$

- (i) $\beta = 0.99$: $\gamma = 7.089 \implies B \ll 4.398 \times 10^{13} \text{ Gauss}$
- (ii) $\beta = 0.999$: $\gamma = 22.366 \implies B \ll 4.339 \times 10^{12} \text{ Gauss}$
- (iii) $\beta = 0.1$: $\gamma = 1.005 \implies B \ll 2.144 \times 10^{17} \text{ Gauss}$

Problem 5

Cyclotron Emission Spectrum

Non-relativistic ($\beta \ll 1$) particles accelerated by a magnetic field produce an electric field that evolves purely sinusoidally as seen by an observer. Hence, it's power spectrum only consists of a **single, sharp line** at the fundamental gyration frequency ω_B . The gyration frequency for the non-relativistic electron: $\beta = 0.01$, $B = 10^8$ Gauss is given by:

$$\omega_B = \frac{eB}{\gamma mc} = 1.759 \times 10^{15} \text{ rads}^{-1} \quad (15)$$

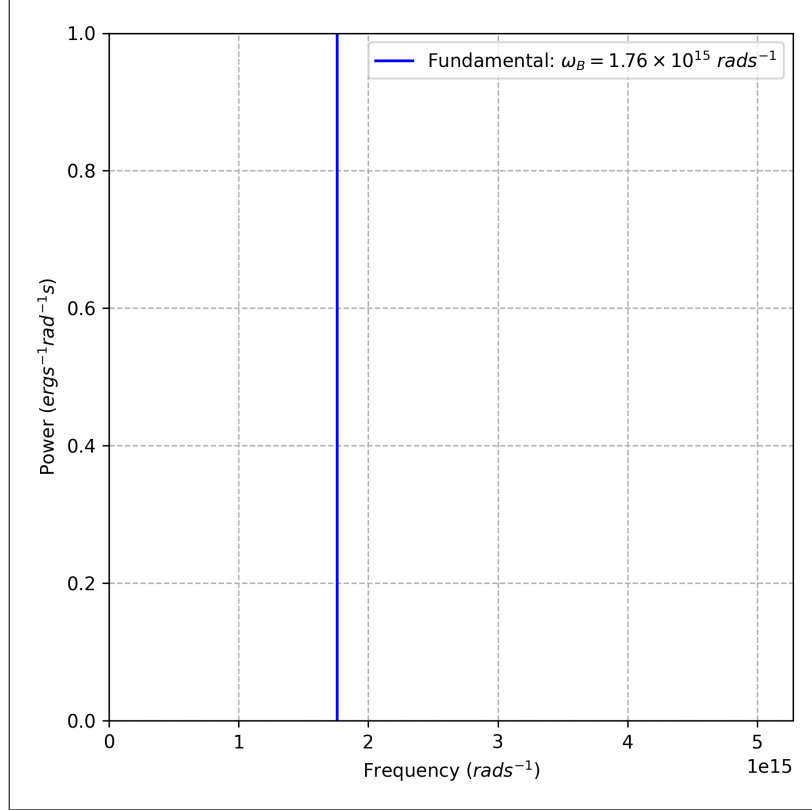


Figure 7: Cyclotron power spectrum

Synchrotron Emission Spectrum

Relativistic ($\beta \rightarrow 1$) particles accelerated by a magnetic field emit radiation that is relativistically beamed into a narrow cone of half-angle $\sim 1/\gamma$. As a result, they produce an electric field that, as seen by an observer, evolves in the form of sharp pulses of width much smaller than the gyration period and time period equal to the gyration period. Therefore, it's power spectrum is a superposition of a large number of harmonics of ω_B with strengths proportional to increasing powers of β , thereby extending to a **broad continuum**. Consequently, the emission power spectra spans to very high frequencies and peaks at a critical frequency ω_c much higher than the gyration frequency. The gyration frequency and the critical frequency for the relativistic electron: $\beta = 0.999$, $B = 10^8$ Gauss, $\alpha = \pi/2$ (assumed) are given by:

$$\omega_B = \frac{eB}{\gamma mc} = 7.864 \times 10^{13} \text{ rads}^{-1} \quad (16)$$

$$\omega_c = \frac{3}{2} \gamma^3 \omega_B \sin \alpha = 1.320 \times 10^{18} \text{ rads}^{-1} \quad (17)$$

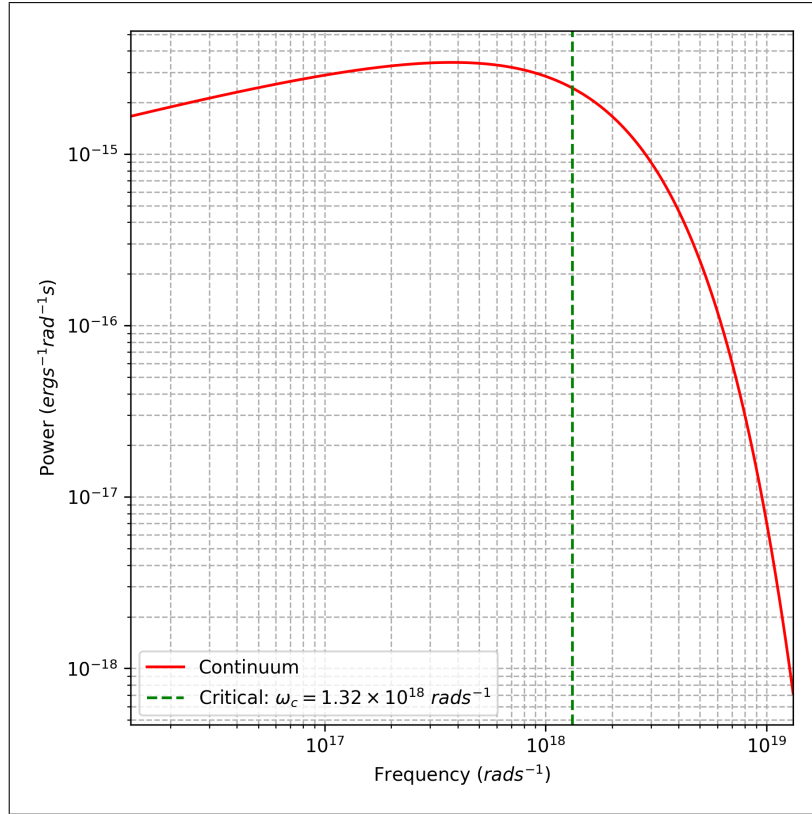


Figure 8: Synchrotron power spectrum

Links to code: [Question 5\(a\)](#), [Question 5\(b\)](#)

Problem 6

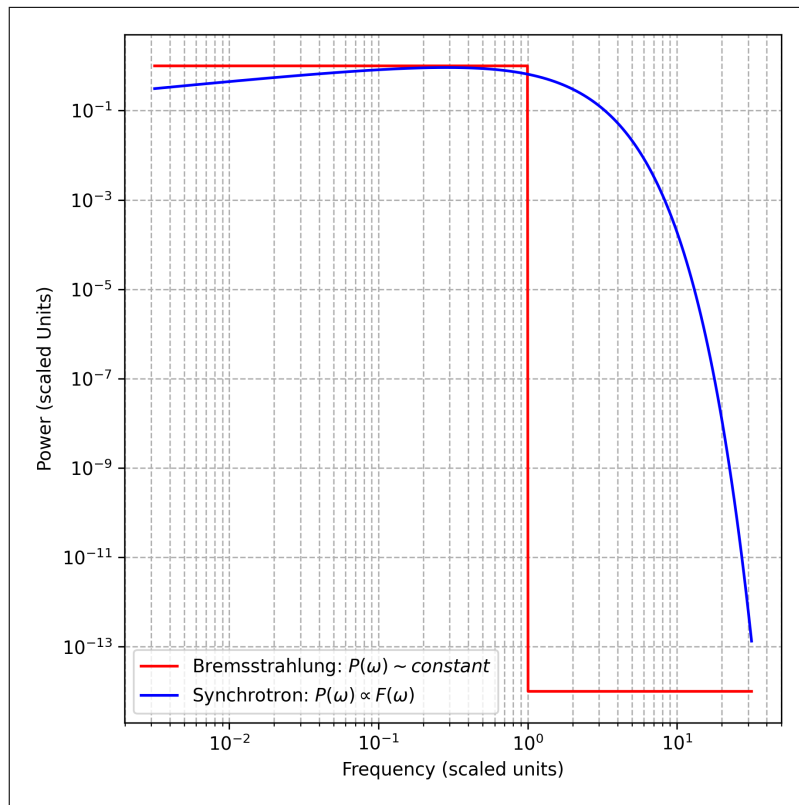


Figure 9: Singular Bremsstrahlung and Synchrotron emission spectra (appropriately scaled)

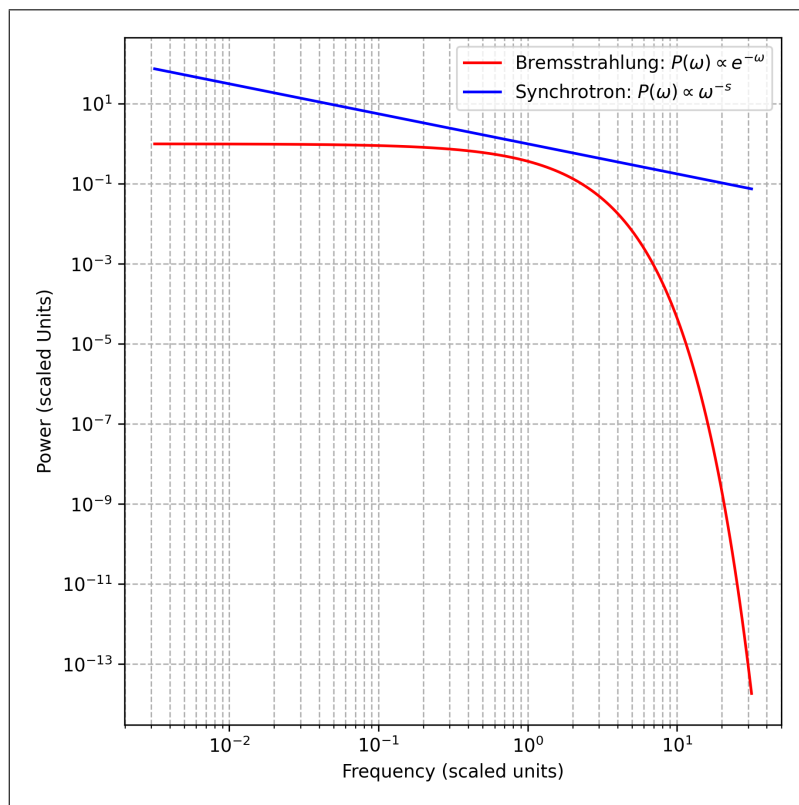


Figure 10: Complete Bremsstrahlung and Synchrotron emission spectra (appropriately scaled)

Bremsstrahlung

Single-particle spectrum: For a single mildly relativistic electron, the spectrum is almost flat at low frequencies with a sharp cut-off at high frequencies defined by the kinematic limit ($h\nu < \frac{1}{2}mv^2$).

Multi-particle spectrum: For a thermal (Maxwellian) distribution of electrons, the sharp cut-off is smoothed out, and the spectrum is almost flat at low frequencies with an exponential tail ($e^{-h\nu/kT}$) at high frequencies.

The key physical reasons for the difference in spectral properties are

Emission Mechanism: Bremsstrahlung is a free-free emission arising from the acceleration of free particles by discrete collisions with free ions.

Kinematic Regime: Bremsstrahlung is produced by mildly relativistic particles with no localised pulsing effects.

Plasma Nature: Bremsstrahlung is typically a thermal process governed by the temperature of the Maxwellian distribution of particle velocities.

Links to code: [Question 6\(i\)](#), [Question 6\(ii\)](#)

Synchrotron Emission

Single-particle spectrum: For a single ultra-relativistic electron, due to relativistic beaming, the spectrum is a continuum of harmonics ($F(\omega)$) with a peak around the critical frequency.

Multi-particle spectrum: For a non-thermal power law distribution (γ^{-p}) of electrons, the spectral function evaluates to a constant, and the spectrum is also a power law continuum (ω^{-s} , $s = (p - 1)/2$).

The key physical reasons for the difference in spectral properties are

Emission Mechanism: Synchrotron emission is a bound emission arising from the acceleration of particles in continuous bound helical orbits in a magnetic field.

Kinematic Regime: Synchrotron emission is produced by ultra-relativistic particles with strong relativistic beaming.

Plasma Nature: Synchrotron emission is typically a non-thermal process governed by the power law distribution of the particle energies.